



Characterizing E-Commerce Harm by Investigating Online Communities

A Case Study With Abusive Dropshipping

Arjun Arunasalam¹ and Z. Berkay Celik² | Purdue University

E-commerce websites are targets of abusive individuals, though it is difficult to understand the methods and tools these individuals employ. We introduce a methodology and case study that leverage online communities as data sources to identify and analyze harmful activities.

The growth of e-commerce platforms, including Amazon, eBay, and Shopify, has led to a diverse range of users participating in this online marketplace. Users create accounts on these platforms to purchase items as customers. Simultaneously, users can also establish business accounts to become sellers, offering goods to customers through a digital storefront. Customers benefit from the convenience of e-commerce shopping, while sellers often find success in operating an online business. In this environment, both buyers and sellers are expected to adhere to platform guidelines, and they often do. However, not all users comply.

Some individuals—abusers—engage in harmful activities on e-commerce platforms, employing methods that violate platform guidelines. However, identifying and understanding previously unexplored forms of harm, such as the tactics these individuals use to violate guidelines and the third-party software and tools they employ, remains challenging. Through a case study of abusive dropshippers¹ (individuals who exploit the legitimate dropshipping business model), we demonstrate how qualitative analysis of underground online

communities can serve as a valuable information source to address this challenge.

Abusive dropshippers take advantage of other legitimate sellers to resell items (detailed below), causing harm to users, legitimate sellers, and the platform itself. We present a review of key findings, including third-party web extensions used by the community; guidelines for forged documents; and web services for falsifying tracking numbers. Our findings were extracted via a combination of inductive and deductive analyses of forum threads from online communities, demonstrating their effectiveness in characterizing harm. We conclude by emphasizing the importance of using online communities as a data source for e-commerce stakeholders to understand the characteristics of harm and minimize it.

Monetizing via E-Commerce

E-commerce platforms have experienced significant growth in recent years, with a plethora of new online shopping websites emerging regularly (e.g., Temu and Shein). These platforms function as sociotechnical systems, serving as digital interfaces where customers engage in transactions with sellers. Customers appreciate the convenience of buying goods from home, while sellers

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are drawn to the low barrier of entry and ease of operation. Anyone can open an e-commerce storefront on a third-party website by meeting minimal requirements, such as providing their name and business location. This, coupled with factors such as reaching a wide audience and the ability to run a store remotely from anywhere in the world, has popularized e-commerce as a means of monetization. As members of this sociotechnical ecosystem, both customers and sellers are expected to act responsibly and adhere to platform guidelines to ensure a safe and mutually beneficial e-commerce environment.

(Ab)users Who Violate Guidelines

It is important to acknowledge that not all participants in online marketplaces act responsibly. Some individuals, referred to as *abusers*, engage in harmful activities on e-commerce platforms, employing methods that violate platform guidelines. For example, customers might falsely claim that they have not received an item to get an e-commerce store to send another item (getting two for the price of one). Similarly, sellers might engage in prohibited activities, such as soliciting fake reviews.

Here, we note that abuse on e-commerce platforms takes many forms, with both customers and sellers using tactics that range from exploiting existing features (e.g., free returns) to using third-party tools to automate harmful activities [e.g., artificial intelligence (AI)-generated reviews].

Challenge of Characterizing Harm

Given the diverse abusive tactics used by individuals on e-commerce platforms, it is vital that these platforms accurately identify and understand the harm being caused, including the specific tools and tactics used. This understanding is crucial because it reveals weaknesses in a platform's defenses, which impedes the development of effective countermeasures.

However, despite its importance, characterizing harm comes with a plethora of challenges. First, platforms may not be aware of the abuse taking place because abusers often operate in secret. Many platforms try to limit problematic behavior. For example, an e-commerce platform might notice if someone creates several seller accounts from a single Internet Protocol address and ban that user. Because of this, abusers often take extra steps to avoid being detected and appear legitimate. They can devise strategies to mimic normal or conventional accounts. For example, someone who purchases fake reviews for their e-commerce profile from a third-party service may also solicit reviews from real users (e.g., by offering incentives, such as free or heavily discounted items).

Second, abusers are constantly adapting their methods to avoid detection. Consider a platform that implements an algorithm or machine learning (ML) model to detect fake reviews. Once abusers realize this, they may

decide to switch to a different third-party review service that uses more advanced methods. For example, they might switch to a service that uses large language models (LLMs) to produce reviews that achieve satisfactory perplexity scores (measurements of how well a model predicts the next word in a sequence), a metric used to distinguish between bots and humans.

Third, harm may not be limited to a single platform. Consider an abuser who engages in harmful activities on two different e-commerce platforms. The abuser might purchase from platform A and use those products to fulfill orders on platform B (detailed in our case study below), an act prohibited by most e-commerce platforms. The lack of communication between the two platforms means that neither one has the full picture of the harm being caused to stakeholders in the platform.

Online Communities as Data

We propose that online communities can be a valuable resource for identifying and understanding harmful activities. Online communities, such as Reddit and 4chan, have become major platforms where people engage in various types of discussions. One type of discussion involves community learning, in which people learn from one another. This learning is not always positive; individuals seeking to exploit or misuse services also often seek information in these communities.

In addition to mainstream channels, such as Reddit and 4chan, discussions can also occur in more specialized and niche online communities. These range from communities that require moderator approval for new members to join to forums that are language specific (e.g., Russian, Turkish, and Vietnamese).

In both mainstream and specialized channels, individuals engaging in harmful activities not only discuss tactics to carry out abuse but also share links to external resources (e.g., tools, software, and guidelines). This makes online communities a valuable source of data for understanding and characterizing harm.

Recognizing the value of online communities as a data source, we outline a methodology to characterize harm using information from these communities. We draw from social science frameworks, primarily digital ethnography, to understand online communities of abusers. To uncover abuse on e-commerce platforms, we propose a three-stage process: 1) community discovery, 2) data analysis to extract findings, and 3) triangulation to confirm findings.

Community Discovery

The first step is to identify communities where abusers discuss their activities. These are primarily forums dedicated to specific types of abuse or forums that host subcommunities focused on such topics (e.g., a dedicated subreddit on Reddit).

To find these communities, we begin with a set of seed keywords. These keywords can be general (e.g., “e-commerce abuse”) or more specific to a particular type of abuse (“buy fake reviews”). These keywords serve as an entry point for searching the web using a simple query crawler, such as one that returns the top results from a search engine. To improve the quality of the data collection, we can expand this set of keywords. For instance, we can find related keywords using the Google Trends application programming interface (API) or use search engine optimization (SEO) APIs that provide keywords related to a specific word or theme.

These keywords give us a list of uniform resource locators (URLs) that may or may not lead to relevant forums. We can then use a filtering mechanism, such as regular-expression matching (patterns like “r/” and “forum.” can indicate online communities) or ML methods (e.g., LLM as a judge), to identify a list of candidate online communities for further analysis.

Data Analysis to Extract Findings

Once we have identified potential online communities, the next step is to analyze the data from these communities. The data primarily consist of forum threads, which typically include an initial post and a discussion thread where users interact by adding comments. These comments can contain valuable information about the tactics, tools, and motivations of abusers.

Should we analyze all threads? Not necessarily. An additional filtering step can be helpful. We can use various methods, from keyword search to a Bidirectional Encoder Representations From Transformers (BERT)-based classifier (trained on manually annotated threads), to improve the data quality and select relevant threads for further analysis. For example, in the case of keyword search, if we are interested in understanding how abusers falsify tracking numbers, we might use keywords such as “tracking,” “fake,” and “delivery” to filter the threads. In the case of BERT, which is an ML model that excels at understanding the context and meaning of words in a sentence, we can fine-tune a BERT model to classify forum threads as relevant or irrelevant based on their content, helping us focus our analysis on the most pertinent discussions. Here, additional considerations can be made. For example, we might use a loss function that penalizes false negatives more than false positives, as our objective is to avoid missing pertinent threads.

Qualitative analysis is a suitable method for analyzing the filtered threads. This approach involves a combination of inductive and deductive reasoning to identify patterns and themes related to the specific type of abuse we are investigating. For instance, if we are studying abusive dropshippers, we might uncover themes such as “methods for evading platform detection,” “tools

for automating dropshipping processes,” or “strategies for obtaining counterfeit goods.” In practice, multiple researchers can work together to iteratively sample threads, identify recurring patterns or concepts (low-level themes), and group these into broader categories (high-level themes). This process leads to the creation of a codebook, a guide that defines and explains the identified themes. The iterative sampling continues until no new themes emerge from the analysis, indicating that the codebook is comprehensive. We can systematically apply a stable codebook to the remaining threads to gain a deeper understanding of the abusive behaviors prevalent in the online community.

Data Triangulation

The themes identified via qualitative analysis are valuable for characterizing harm and showcasing the tools and tactics abusers use. From this characterization, we can understand how abusers impact victims and stakeholders. Yet, there is additional value in triangulation, the process of using additional data sources or methods to validate and strengthen our findings.

There are various options for triangulation. For example, if we know that abusers who create fake reviews harm unsuspecting customers, it is useful to gather information from people who unknowingly bought an item based on fake reviews. Similarly, if an abusive seller harms other sellers, it is valuable to collect data from the victimized sellers (e.g., through interviews and surveys). Other potential triangulation methods include analyzing platform data logs or conducting experiments to test the effectiveness of identified abusive tactics.

Triangulation is helpful, as it can not only confirm findings from qualitative analysis but also provide new insights. By incorporating data from different sources, we can address the potential limitations of relying solely on data from online communities, such as bias or incomplete information. Furthermore, triangulation can help us understand the real-world impact of the abusive behaviors identified in online communities.

Case Study: Abusive Dropshippers

We now present a case study where we examine abusive dropshippers, individuals who engage in a harmful form of reselling that is strictly prohibited by e-commerce platforms. To understand this community, we use our three-stage pipeline described previously.

What Is Dropshipping?

Dropshipping is a retail business model where sellers manage customer orders without maintaining inventory. Sellers, known as *dropshippers*, establish agreements with source vendors to fulfill orders on their behalf. Dropshipping can be performed on various e-commerce

platforms, such as Shopify, eBay, and Amazon, each with their own advantages and disadvantages. Dropshippers can select their preferred platform per individual needs. For example, Shopify has lower transaction fees but attracts less consumer traffic than Amazon. Dropshippers select items to sell based on market research, personal interests, or current availability.

As with most business models, there are guidelines for dropshipping. Compliant dropshippers adhere to platform regulations, ensuring that their operations align with these guidelines. Primarily, dropshippers establish formal agreements with vendors, guaranteeing product availability and ensuring timely fulfillment of customer orders. The agreement outlines terms by which the vendor handles order fulfillment and the dropshipper assumes responsibility for customer interactions, including returns and refunds. Both parties also determine the percentage cut the vendor receives for each item sold by the dropshipper.

After setting up their store, dropshippers fulfill customer orders in one of two ways: through the source vendor, who ships directly to the customer using the dropshipper's information, or by using third-party fulfillment services that store and ship items. These services, such as Amazon's Fulfilled by Amazon (FBA), streamline the process by centralizing product returns and inventory management.

What Is Abusive Dropshipping?

Figure 1 provides an overview of the abusive dropshipping process. An abusive dropshipper imitates legitimate dropshipping but exploits the process by secretly sourcing items without proper agreements. They operate by selecting two e-commerce platform domains: the source domain, where they buy items from private sellers, and the target domain, where they list these items for sale. Unlike legitimate dropshippers, abusive dropshippers do not establish formal agreements with

the source vendors. Instead, they list items from private sellers on the target domain without the sellers' consent or knowledge, violating the regulations of most e-commerce platforms.

To fulfill customer orders, the abusive dropshipper purchases the item from the private seller, using the customer's shipping information. For instance, an abusive dropshipper might list a pair of shoes for US\$50 on the target domain. When a customer places an order, the dropshipper buys the same shoes for US\$30 from a private seller on the source domain, using the customer's address for delivery. The private seller then ships the item using their own label, unaware that the order was placed by the abusive dropshipper and not the customer. Customers, in turn, are unaware that their orders are being fulfilled by another private seller.

Abusive dropshipping operations violate several key e-commerce guidelines, including requirements for supplier agreements, proper shipping and packaging, and stock control. These violations classify their operations as abusive, similar to other practices that violate online platform guidelines, such as toxic content, misinformation, and search rank manipulation.

Research Questions

Our study of abusive dropshipping focuses on these three research questions:

- RQ1: What methods do abusive dropshippers use that violate e-commerce policies?
- RQ2: What tools are used to carry out these methods?
- RQ3: How do these abusive operations harm customers, other sellers, or e-commerce platforms?

Executing Our Three-Stage Pipeline

Abusive Dropshipper Communities. We began with an initial crawl of 238 unique webpages using Google search queries, such as "dropshipping" and "dropship."

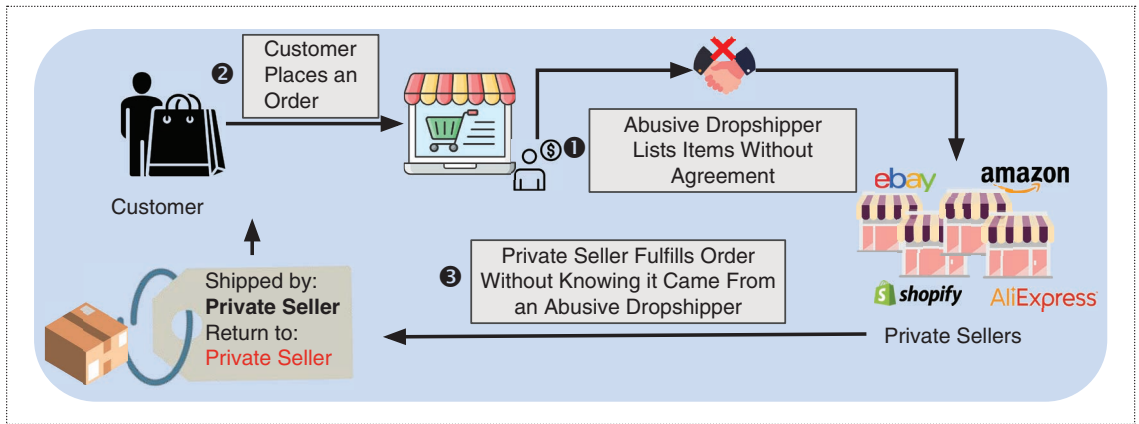


Figure 1. An overview of abusive dropshipping.

Only 24% of these pages discussed abusive dropshipping. To find more abusive content, we added 24 more specific queries, such as “dropshipping without permission,” which led to the discovery of 931 additional webpages. A random sample showed that 82% of these pages contained relevant dropshipping content.

We then grouped the pages by top-level domain and found that 40% discussed abusive dropshipping, with 70% of this content coming from discussion forums. The study focused on seven popular forums: two dedicated entirely to dropshipping and others with specific subgroups that discussed it. Interestingly, forums in non-English languages (Turkish, Vietnamese, and Indonesian) were also found, indicating that abusive dropshipping practices are global. We crawled these forums again, this time using translated queries, eventually collecting 3,651 threads discussing abusive dropshipping.

Qualitative Analysis. We conducted a thematic analysis on 1,050 threads (30% of the total) after iteratively sampling 210 threads from the seven forums. Approximately 70% of the threads involved discussions about abusive dropshipping, while 25% included complaints from private sellers who were exploited. Only 5% of the threads were related to compliant dropshipping.

During the thematic analysis, we discovered two additional external resources: instructional materials and software tools. The instructional materials consisted of four widely used online guides, including text-based and video content. We also analyzed software tools used by abusive dropshippers, including benign tools but also tools with solely malicious capabilities.

Triangulation to Understand Seller Harm. To further understand the impact of abusive dropshipping on customers, sellers, and e-commerce platforms, we conducted semistructured interviews with six qualified individuals, including four FBA sellers, one compliant dropshipper, and one legal consultant. The recruitment involved purposive and snowball sampling, starting with 11 potential participants, with only six passing the screening process. Our study was reviewed by analysts from our institutional review board, who ensured that our protocol adhered to ethical guidelines. Our study was granted exemption status.

Findings

We present a selection of findings from our three-stage pipeline. First, we briefly introduce the different tactics and behaviors in which abusive dropshippers engage. Second, we outline abusive dropshippers’ use of software. Finally, we demonstrate how abusive dropshippers are marked by specific characteristics, which can be helpful for identifying and detecting these individuals.

Abusive Dropshipping Tactics

Broadly, the themes we discovered that relate to tactics pertain to how abusive dropshippers 1) target specific items and 2) engage in covert behavior, such as concealing tracking information and document forgery.

Targeting Gap Items. Abusive dropshippers take advantage of “gap items,” products either unavailable or priced higher on their target platform compared to their source platform. We identified two motivations for gap item selection: exclusivity and competitive profit.

Exclusivity-motivated gap items are those that are unavailable on the target platform but are desired by customers, such as U.S. products sought by consumers in Canada and Mexico or Japanese products desired by Japanophile communities in Europe. Abusive dropshippers exploit this demand by marking up prices significantly, sometimes by three times. This practice not only burdens consumers with unfair pricing but also risks damaging the reputation and brand image of the original source merchants. Customers, expecting higher quality due to the inflated price, often leave negative reviews when a product does not meet their expectations, further harming the brand’s reputation. Here, it is important to note that negative reviews on a product can adversely affect the reputation of any seller who offers that product.

Competitive profit-motivated gap items are those that abusive dropshippers can sell at a lower price than existing listings on the target platform. For instance, a baby bottle might be priced at US\$15 on Amazon Mexico but sourced for US\$7 from Amazon U.S. The dropshipper then lists the item for US\$12, earning a US\$5 profit per sale. This strategy allows them to generate substantial revenue, as illustrated by one forum user who was an abusive dropshipper: “In a month I was able to bank 10K ... a 40% margin, so pretty good numbers.”

Avoiding Trademarked Items. Another interesting aspect of our findings is the strategic avoidance of trademarked items. Abusive dropshippers are keenly aware that listing items from well-known brands, such as Adidas or Nike, can lead to swift action from these companies, resulting in the removal of their listings. Instead, they focus on “nonrisky items,” which are from brands less likely to file complaints. These items are identified through collaborative efforts within dropshipping communities, where users maintain and share lists of such brands to avoid detection and suspension from e-commerce platforms.

These collaborations extend to creating aggregated lists of profitable gap items, often referred to as “winning products” within these communities. Abusive dropshippers frequently request access to these lists, which contain thousands of items proven

to sell well in specific markets, such as items sourced from the United States to be sold in Mexico. The persistence of these operations is further ensured by updating these lists based on experiences with brand complaints, thereby helping other dropshippers avoid problematic items.

Document Forgery. Interestingly, when faced with brand complaints, some dropshippers resort to forging “Letter of Authorization” to prove they have the right to sell certain items. This tactic, supported by instructional materials and forum advice, allows them to bypass e-commerce platform regulations, thereby continuing their operations despite legal challenges. To illustrate, one video we analyzed had a section titled “How to Circumvent License of Authorization on Amazon (Sample License of Authorization Text),” with a sample document to provide to an e-commerce platform.

Concealing Shipping Information. Abusive dropshippers often withhold shipment tracking information from customers for two primary reasons. First, providing tracking numbers from private sellers can lead to account suspension on e-commerce platforms. This occurs when platforms detect a discrepancy between the listed address of the dropshipper and the actual shipping origin, which often comes from a different region or country. Second, abusive dropshippers fear that revealing the foreign origin of a shipment may raise customer suspicion, leading to a loss of credibility and reputation. For example, if a customer in Mexico sees that their order is coming from another country, they may doubt the legitimacy of the purchase.

Thus, abusive dropshippers often exclude tracking information. This lack of transparency is detrimental to customers, who are left in the dark about the status of their shipments. Unlike legitimate sellers, who have no reason to withhold tracking information, abusive dropshippers deliberately obscure these data, causing significant problems. An interviewee (from our triangulation efforts) noted that the absence of tracking numbers is particularly problematic for international shipments, where multiple transfer points increase the risk of damage. The lack of transparency leaves customers uncertain about when or if their product will arrive, damaging trust and satisfaction.

Use of Software Tools

Beyond just behavior, we also discovered the exploitative use of benign and malicious software tools to aid abusive dropshippers’ operation.

Benign Tools. Abusive dropshippers exploit product research software to identify profitable gap items, using

geographic filtering and broad keywords. This software automates queries across e-commerce platforms, helping dropshippers find items from multiple private sellers. Abusive dropshippers also use repricing tools to ensure that their products are competitively priced by automatically adjusting prices based on market comparisons. Additionally, abusive dropshippers use review software to solicit positive feedback, prompting customers to leave five-star reviews. These strategies help them maintain competitive pricing, attract more customers, and enhance the perceived value of their products, even when operating unethically.

Exploitative Web Services. Abusive dropshippers often generate fake tracking numbers using services such as BlueCare Express and PackTrack to deceive customers and e-commerce platforms. These fake numbers allow abusive dropshippers to manipulate order statuses, such as “shipped” or “delivered,” even if the items have not yet been sent (since the dropshippers depend on the exploited seller to send the product).

In some cases, abusive dropshippers use real tracking numbers associated with existing unrelated orders. As seen in Figure 2, they use web tools that allow them to provide criteria such as “delivery by” dates and destination postal codes to retrieve tracking numbers that meet these criteria. This practice misleads customers, who might receive incorrect tracking details.

Community-Designed Malicious Software. Software use does not stop with exploiting existing tools. We discovered that this community also designs software specifically for abusive purposes. To demonstrate how malicious software can be exploited to find and avoid trademarked items, we analyzed the execution of a web extension tool built and shared among abusive dropshippers. This tool, marketed as product research and listing software, integrates into a browser to quickly access and upload product lists with a single click. By

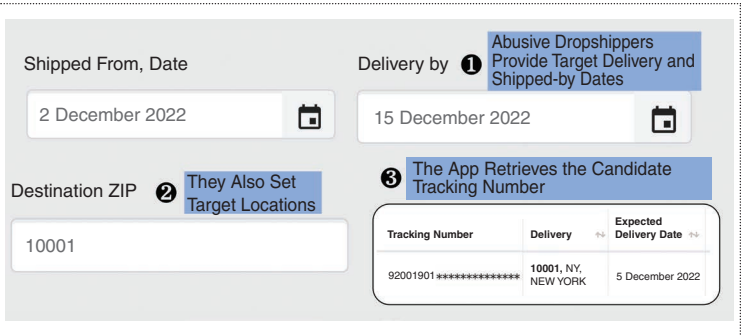


Figure 2. A web tool that abusive dropshippers can leverage to retrieve candidate tracking numbers to provide customers.

installing and examining it using Chrome's DevTools, we discovered that the tool crawls item pages based on user-submitted queries, collecting details, such as the item name, brand, price, and shipping information. It also filters out brands listed in a denylist maintained by abusive dropshippers, which includes brands that have filed complaints against them.

The tool further enhances its capabilities by checking trademark statuses through public services, such as Bulk-SEO-Tools and Trademarkia. Items from trademarked brands are automatically rejected, allowing dropshippers to avoid potential legal issues and continue their operations. The tool also minimizes competition by checking whether an item is already listed by another seller on the target platform, using its Amazon Standard Identification Number. The filtered results are saved in a comma-separated value file for further inspection. This functionality helps abusive dropshippers maintain longevity by circumventing platform regulations and avoiding competition from established brands.

Identifiable Features for Abusive Dropshippers

Aside from exposing tactics and behaviors, our findings help inform how abusive dropshipping can be identified by five key traits aimed at ensuring operation longevity, maximizing convenience and profit, and minimizing losses, as reported in Table 1. These characteristics are observable by customers through public information on a seller's online storefront.

Diverse Item Categories and Brands (A1-A3). Abusive dropshippers typically list items across many categories and brands (A1) but have few listings per brand (A2). They often use software to filter out trademarked

items (A3) to avoid account suspension due to infringement complaints. This strategy allows them to cater to a wider range of customers and limit the impact of platform delistings to specific items. In contrast, legitimate sellers usually focus on niche products, listing many items under a few selected brands.

Geographic Location (A4). Abusive dropshippers often operate from countries different from the marketplaces they sell on, targeting specific consumer bases where there is minimal competition. This practice is highlighted in instructional materials, which suggest that selling in regions with less saturated markets increases sales. Legitimate sellers, however, typically have a local business address to manage shipping and returns.

Low-Priced Items (A5). Abusive dropshippers list a high number of items at low prices to reduce the risk of trademark complaints and minimize losses if returns are requested. Lower-priced items are less likely to be returned, and even if they are, the financial loss is minimal, as these dropshippers often allow customers to keep the item rather than processing returns.

Minimizing Stakeholder Harm Nature of Harm

E-commerce abuse, such as abusive dropshipping, causes online harm, which can range from unethical behavior to online crime, depending on the tactics used. In the case of abusive dropshipping, deceptive reselling itself violates platform policies, which, in itself, is not a crime. However, practices such as forging supplier agreements and generating fake tracking numbers can constitute criminal offenses under certain consumer protection and fraud laws [e.g., those put in place by the U.S. Federal Trade Commission (FTC)]. Additionally, in certain circumstances, dropshippers may be violating intellectual property laws when selling items they did not acquire the rights to sell (echoed by our interviews with legal consultants).

Regardless, the harm that stems from e-commerce abuse does impact customers. To illustrate, when customers attempt a return/refund, they may attempt to return the item to the exploited seller who unknowingly fulfilled the order. This seller may be unwilling to process the return/refund, as they have no record of transaction with the real customer (only the abusive dropshipper). This can lead to denied refunds or prolonged disputes. In some cases, it can lead to the customer's account being banned if the exploited seller flags the customer for supposed return abuse. A similar process follows if the customer attempts to return the item to the abusive dropshipper. Here,

Table 1. The five traits that serve as identifiable characteristics of abusive dropshippers.

Feature	Description
Diverse item categories (A1)	Listing items across many categories
Few listings per brand (A2)	Despite diversity of brands, each brand has only a small number of listings
Brands offered (A3)	Target nontrademarked items
Geographic location (A4)	Operate from a different country than the target platform
Low-priced items (A5)	Items selected for listing are on lower end of prices

the abusive dropshipper may also maliciously prolong disputes and flag the customer to avoid having to refund them.

Additionally, when the abusive dropshipper is unable to procure an exact match from a seller, customers may receive items that do not match their description. Moreover, practices such as claiming nonexistent inventory and issuing fake tracking numbers—which may result in buyers not receiving items at all—can have long-term effects on user trust in an e-commerce platform, resulting in customers favoring other platforms.

Tackling E-Commerce Abuse

We argue that stakeholders within e-commerce platforms should use our insights to reduce harm. We envision this happening in two ways: first, through platform detection and prevention of abusive activities and second, by increasing awareness among legitimate users.

In our case study, we show how our analysis allows us to identify abusive dropshippers based on a set of characteristics. This suggests that these characteristics can be leveraged by e-commerce platforms as a way to monitor sellers that match these characteristics. By monitoring said sellers, it is possible that online platforms can get deeper insights into this community of abusive sellers. For example, given that e-commerce platforms have full insight into data surrounding all sellers, flagged sellers can be monitored to verify how often they list or delist items. This allows platforms to build on our discovered characteristics.

In addition, platforms should provide better support for customers and sellers who become victims of harm. Although users facing abuse, such as toxic content, often have reporting mechanisms available, such mechanisms are either lacking or ineffective for e-commerce abuse.

In our study, the sellers interviewed expressed difficulty in reporting abusive practices of other sellers. Their complaints often go unheard (most were unable to reach a human, often interacting only with a bot after filing a complaint). Platforms should simplify the process of reporting abuse for both sellers and customers to help resolve disputes between two parties (e.g., verifying the alleged abusive dropshipper's letter of authorization when a victim seller has the sole authorization to sell an item within a region).

It is important to note that because abuse is not confined to a single platform, an abusive dropshipper can victimize a seller on Etsy to sell on Amazon. Therefore, some level of cross collaboration among platforms is necessary for effective abuse prevention. Future work should explore the feasibility of such efforts.

Going Forward

Our case study demonstrates the wealth of information available through our three-stage pipeline. It is crucial to continue monitoring online communities in the future.

The Lowering Barrier of Entry to Abuse

E-commerce abuse has a low barrier of entry. Anyone can create an account to sell/buy and then engage in harmful behavior while remaining undetected. Additionally, online communities have become a learning ground for aspiring abusers.

We argue that the already low barrier of entry will continue to erode, especially with the emergence of new publicly accessible technology. For example, LLMs have not only inspired a plethora of AI agents for beneficial purposes but also for malicious ones.² It has become possible for individuals with limited technical skills to design harmful tools powered by these models. Moreover, these tools can be rapidly updated to improve their performance while maintaining covert behavior. The easy access to powerful technology for end users can worsen the problem of abuse on e-commerce platforms.

Exploring More Hidden Communities

Our methodology focuses on discoverable online communities accessible to anyone on the Internet (their forum endpoints can be accessed by anyone with a web browser). Through our case study, it is clear that even these online communities provide a wealth of information. However, it is important to note that these findings represent only a portion of abusive e-commerce behavior. In fact, more activity might be discussed in other hidden online communities. For example, communities that exist on the dark web are likely to discuss more serious cases of abuse.

Thus, future work should consider including these communities in its analysis pipeline. Additionally, communities may not exist solely on web forums. Platforms such as Telegram are known to host a variety of online communities, some of which may be abusive, making the platforms a valuable source of information. However, it is crucial to continue considering the ethics of studying more hidden communities. People in these spaces intentionally conceal themselves to avoid being observed. This is also the case for individuals who join invite-only Telegram groups. To what extent should researchers go to gain access to private groups to analyze data? The answer to this question often elicits diverse and conflicting opinions among researchers. Furthermore, most of the harm we uncover is abusive and exploitative since it violates the terms of service and guidelines of the platform. More serious abuse may cross into illegal activity. Researchers who discover such activity may need to consider developing

guidelines for reporting illegal activity to appropriate law enforcement agencies. What if illegal activity spans across different regions? Such questions need to be carefully addressed in study designs that involve more hidden communities.

Stakeholder Roles in Combating Abuse

E-commerce Platforms' Reactions. It is also important to note that as primary stakeholders, e-commerce platforms themselves should partake in efforts to monitor underground communities of abusers, as they stand to benefit from such monitoring. For example, platforms can proactively scrape targeted forums to collect references and URLs to third-party web applications, software, and services used by abusive actors. By doing so, they can gain greater visibility over the evolving tactics, tools, and discussion patterns within these underground groups. Beyond passive monitoring, platforms can also leverage natural language processing and entity recognition to flag recurring patterns in forum discussions. To illustrate, by identifying frequently mentioned services (e.g., fake tracking number generators and forged authorization letter providers), platforms can compile internal watch lists of emerging threats and tools.

Researcher-Platform Collaboration. Beyond gaining insight, e-commerce platforms should collaborate with researchers to detect abusers who violate guidelines. Given our discovered traits, it is straightforward to build a classification model so long as there is a labeled dataset of "normal" sellers and threat actors (e.g., abusive dropshippers). Acquiring such a dataset would require knowledge sharing from e-commerce platforms. To realize such a classification model, we propose a collaboration between researchers who investigate specific forms of abuse and relevant e-commerce platforms. For instance, by leveraging our discovered traits, we can collect data from vetted e-commerce business (e.g., FBA) and leverage an anomaly detection model to discover candidate e-commerce threat actors. Providing a list of candidates for the abusive dropshippers to verify can open channels for a long-term collaboration that involves access to datasets for subsequent training of a more robust classifier. Researcher-platform collaboration is essential to build long-term solutions to e-commerce abuse.

Policy Responses to Abuse. Abuse on e-commerce platforms, such as abusive dropshipping, not only harms the platform but erodes consumer trust. Thus, official agencies, such as the FTC, bear a responsibility in taking a regulatory approach. For instance, mandating that sellers report verified supplier agreements and shipping sources is one potential approach. The FTC should also

work in tandem with platforms to enforce identity verification (e.g., such as verifying a seller's listed suppliers).

Additionally, policy makers should also hold platforms accountable by enforcing reporting standards and ensuring a clear process for handling consumer complaints. For instance, platforms that are found not to handle customer/e-commerce seller complaints in an orderly manner could be fined. Additionally, the FTC should foster cross-platform collaboration since e-commerce abuse can migrate among platforms. Here, the FTC could collect data on fraudulent accounts, deceptive tools, and other suspicious URLs from each platform; compile them; and circulate such information among participating platforms. The FTC could also take more fine-grained steps, such as having a database of tracking numbers provided to customers in e-commerce transactions. This would allow the flagging of suspicious operations among different platforms (e.g., the reuse of tracking numbers on two different platforms). Through such efforts, policy makers could mitigate the effects of e-commerce online harm.

Complementary Data Sources

Our efforts have focused primarily on analyzing online communities. However, we note that these spaces often provide links to additional resources that exist in online spaces, such as marketplaces for software or collections of video guidelines for abusive activity. These spaces serve as an additional data source, suitable not only for qualitative analysis but also for quantitative insights. For example, consider the abusive behavior of buying and selling e-commerce accounts (to overcome prior suspension). By crawling and analyzing quantitative trends on marketplaces that sell such e-commerce accounts, stakeholders should be able to gather useful information that can aid in detection.

Abuse on e-commerce platforms is common. The low barrier to entry for abuse has resulted in many individuals being willing to exploit platforms for profit. Characterizing the harm that abusers cause and the tools and tactics they use has proved challenging. We proposed online communities as a data source to overcome this challenge. Through a richly informative case study, we demonstrated the value of a three-stage methodology pipeline to discover online communities, analyze data, and triangulate findings.

Going forward, it is essential for e-commerce stakeholders to consider online communities a valuable source of threat intelligence, where they can retrieve deep insights on abuse. By monitoring such spaces, stakeholders can implement defensive and preventive measures to limit the scope and impact of harm on their

platforms. Overall, characterizing harm on e-commerce platforms can help improve this sociotechnical system for the responsible parties (legitimate sellers and buyers) who engage in e-commerce. ■

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Arjun Arunasalam is a Ph.D. candidate in the Department of Computer Science, Purdue University, West Lafayette, IN 47907 USA. His research interests include human factors in security and privacy, with a focus on abuse in sociotechnical systems. Arunasalam received his BSE in computer engineering from the University of Michigan, Ann Arbor. Contact him at aarunasa@purdue.edu.

Z. Berkay Celik is an assistant professor in the Department of Computer Science, Purdue University, West Lafayette, IN 47907 USA, and a codirector of Purdue Security Laboratory. His research interests include the design and evaluation of security in software and systems. Celik received a Ph.D. in computer science and engineering from the Pennsylvania State University. He is a Senior Member of IEEE. Contact him at zcelik@purdue.edu.

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